

AI IN EMBEDDED AUTOMOTIVE SYSTEMS: From Compliance To Intelligent Design

What happens when deterministic embedded workflows meet non-deterministic AI?
The answer is reshaping automotive compliance faster than most realise.



Sarang Savji, Technical Lead, Varroc Engineering Ltd, India, during presentation

The application of artificial intelligence (AI) in embedded systems is emerging as an important area of focus. This is particularly evident in the automotive domain, where complex compliance requirements related to safety, security, and software reliability shape every stage of the development process.

Achieving these compliance requirements involves extensive workflows and process flows that are often time-consuming. At a high level, we can examine how AI and AI-driven tools help simplify these workflows, improve efficiency, and support compliance-oriented

development, while extending their impact across embedded development, manufacturing, and production workflows.

Inside the traditional automotive development workflow

This discussion focuses on the existing embedded development workflow and how AI can be integrated not only into development but also into manufacturing and production workflows. We will briefly examine various compliance requirements, understand the high-level differences among AI, ML, and data analytics, and see how

these principles can improve process efficiency, design optimisation, and model deployment in automotive embedded systems. We will also touch upon sensor data classification, block-level deployment, and the compliance considerations associated with using AI itself.

The traditional embedded workflow typically starts with requirement analysis, brainstorming sessions, and cross-functional discussions to capture every aspect of the product while ensuring compliance. From there, the process moves through architecture design, hardware and software component selection, firmware development,